Extraesophageal Reflux in Children. Chronic Cough, Asthma, the Red Larynx, Salivary Pepsin, and the Aerodigestive Clinic.

Imagine a six-year-old girl who has had a daily, wet-sounding cough for four months. Her voice has become hoarse. An ear, nose and throat surgeon looked at her larynx with a flexible scope and wrote that there was swelling and redness at the back of the larynx, described as posterior laryngitis, consistent with reflux. She has taken omeprazole, a proton pump inhibitor, for eight weeks without any change. Her parents now ask whether the dose should be doubled. Should it?

The main idea of this chapter is that reflux and airway symptoms coexist so often in children that it is easy to mistake association for cause. Gastro-oesophageal reflux, meaning the passage of stomach contents back into the oesophagus, is common in children who cough, wheeze or are hoarse. But in children, giving acid suppression on trust for cough, wheeze, hoarseness or a red-looking larynx does not work, and it carries real harm. The tests that are used outside the oesophagus to suggest reflux, such as the appearance of the larynx, pepsin in saliva and fat-filled cells in lung washings, do not diagnose reflux. When it matters, reflux should be proved with combined pH and impedance monitoring linked to the timing of symptoms. The rest of the effort should go into finding the real cause, which is often a bacterial bronchitis, aspiration during swallowing or a structural problem in the airway. So the answer for this girl is no: the dose should not be doubled. The drug should be stopped and the cough worked up properly.

Some terms first. Extraesophageal reflux means symptoms outside the oesophagus that are attributed to reflux: chronic cough, wheeze and asthma, hoarseness, throat clearing, recurrent pneumonia, and in infants apnoea and noisy breathing. Laryngopharyngeal reflux is a term used mainly by ear, nose and throat surgeons for reflux that is thought to reach the larynx and pharynx. A proton pump inhibitor is a drug that blocks acid production in the stomach; it raises the pH of what refluxes, but it does not reduce how often reflux happens or how much fluid comes back. Multichannel intraluminal impedance with pH monitoring, often shortened to pH-impedance, is a twenty-four-hour test with a thin catheter through the nose that detects the movement of any liquid or gas in the oesophagus, whether acid or not, and measures how high up it travels. Weakly acidic reflux means reflux with a pH between four and seven, which a pH probe alone does not count as reflux.

The evidence in this chapter comes from the 2018 paediatric reflux guideline of the North American and European Societies for Pediatric Gastroenterology, Hepatology and Nutrition, led by Rosen; the 2019 guideline from the American College of Chest Physicians on chronic

cough and reflux in children, led by Chang; the 2020 European Respiratory Society guideline on chronic cough; and the guideline on reflux in children and young people from the National Institute for Health and Care Excellence in England, known as NICE guideline one. Adult perspectives come from the 2022 American College of Gastroenterology guideline, the 2023 American Gastroenterological Association clinical practice update on extraesophageal reflux, and the Lyon Consensus version two, published in 2024. The key trials and studies are the lansoprazole trial in children with poorly controlled asthma published in the Journal of the American Medical Association in 2012, an infant lansoprazole trial from 2009, a Cochrane review updated in 2022, and a series of careful studies from Boston Children's Hospital by Rosen and colleagues on pepsin, laryngeal findings, lung microbiology and aspiration.

Part one. How reflux could reach the airway

There are several ways in which reflux might cause airway symptoms, and it is worth separating them, because each one predicts a different test and a different treatment.
The first is the refluxate itself. What comes back from the stomach is not only acid. It contains pepsin, the protein-digesting enzyme of the stomach, sometimes bile acids, and swallowed bacteria. In children, a large share of reflux is weakly acidic, especially after feeds, and is invisible to a pH probe.
The second is microaspiration. Reflux that reaches the top of the oesophagus, called full-column reflux, can reach the pharynx and be breathed into the larynx or the lungs in small amounts. Pepsin and gastric bacteria found in the airway are, in theory, the footprints of this route.
The third is direct contact with the larynx. Laboratory work by Johnston, Koufman and colleagues, on biopsies from adults with laryngopharyngeal reflux proven by pH monitoring, showed why the larynx might be more fragile than the oesophagus. Carbonic anhydrase three, an enzyme that helps cells neutralise acid at their surface, was absent in sixty-three percent of vocal fold biopsies and severely reduced in another twenty-one percent. E-cadherin, a protein that holds cells together, was partly or completely lost in twenty of fifty-one samples, about thirty-seven percent. Pepsin together with acid caused about two and a third times more tissue damage than acid alone at a pH of two, and about one and two thirds times more at a pH of six. Damage to the fine structure of laryngeal cells began at a pH of four, whereas oesophageal cells were damaged only at a pH of two. Later work showed that laryngeal cells take pepsin inside them by a process called receptor-mediated endocytosis, and that pepsin inside cells damaged mitochondria and the Golgi apparatus even at a neutral pH of seven point four. This is the theory behind non-acid injury. But notice its limits: this is laboratory and adult work. It shows that weakly acidic reflux could plausibly injure the larynx; it does not show that a particular child's hoarseness

is caused by reflux.
The fourth possible route is a reflex. In adults it has been proposed that acid in the lower oesophagus can trigger cough or narrowing of the airways through shared nerve pathways in the vagus nerve. This remains a hypothesis that has been weakly tested in children.
The fifth route runs backwards. Coughing raises the pressure in the abdomen and can push stomach contents up into the oesophagus. So cough can cause reflux. Chang and colleagues studied twenty children with chronic cough using a pH probe together with an objective cough recorder, and analysed five thousand six hundred and twenty-eight coughs. Almost eighty-four percent of coughs were independent of acid reflux. Within a two-minute window, sequences in which cough came first and reflux followed were as common as sequences in which reflux came first, with a median of nineteen per child against twenty-four and a half. Within ten seconds, reflux followed by cough was less frequent than reflux without any cough. The authors concluded that the time relationship was unlikely to be causal.
Why does this matter? Because if the mechanism is microaspiration of non-acid fluid, a proton pump inhibitor, which only changes pH, will not help. If the mechanism is a reflex, acid matters. And if cough is causing reflux, treating reflux cannot stop the cough. Before treating, it is worth deciding which pathway you believe is operating and how you would prove it.

Part two. Chronic cough

The 2019 chest physicians' guideline addressed children aged fourteen or younger with a cough lasting more than four weeks and no known underlying lung disease. It made five recommendations. Do not treat for reflux disease unless there are clinical features of reflux; in other words, cough alone is not a reflux symptom. If clinical features or tests show pathological reflux, treat it according to reflux guidelines, as reflux and not as cough. Do not use acid suppression solely for chronic cough. If the child has gastrointestinal reflux symptoms, treat for four to eight weeks and then reassess, and do not continue proton pump inhibitors or histamine-two receptor blockers beyond that without further evaluation. And investigate according to reflux guidelines. Behind these recommendations is a simple evidence base: the only meta-analysis, which combined two paediatric trials, showed no difference in cough resolution and more serious adverse events with proton pump inhibitors.
The 2022 update of the Cochrane review on reflux treatment for prolonged non-specific cough found six paediatric studies, none of which could be pooled. In an infant trial, the cough outcome actually favoured placebo, with an odds ratio of one point six one and a confidence interval running from zero point five seven to four point five five, which is to say no benefit. Adverse events were clearly more frequent with the proton pump inhibitor, with an odds ratio of five point five six and a confidence interval from one point one eight to

twenty-six point two five. The number needed to harm was eleven over four weeks, meaning that for every eleven infants treated for four weeks, one extra child had an adverse event. The 2020 European Respiratory Society guideline makes a broader point: children are not small adults. Chronic cough in children needs a paediatric algorithm, beginning with whether the cough is wet or dry and looking first for causes such as protracted bacterial bronchitis, which is a persistent bacterial infection of the airways that responds to a course of antibiotics, an inhaled foreign body, or structural airway disease.

Part three. Asthma

Reflux is common in children with asthma. A systematic review by Thakkar and colleagues, published in 2010, gathered nineteen studies including three thousand seven hundred and twenty-six children with asthma and found reported rates of reflux ranging from about nineteen to eighty percent. Much of this reflux is silent.

The decisive trial is the study of lansoprazole in children with poorly controlled asthma, published in 2012 by the American Lung Association Asthma Clinical Research Centers and often called the SARCA trial. Three hundred and six children aged six to seventeen whose asthma was poorly controlled despite inhaled steroids, and who had no symptoms of reflux, were randomly assigned to lansoprazole or placebo for twenty-four weeks. A pH probe was abnormal in forty-nine of the one hundred and fifteen children with an interpretable study, which is forty-three percent, even though none had reflux symptoms. And yet asthma control did not improve. The asthma control questionnaire score changed by minus zero point one with lansoprazole and by minus zero point two with placebo, no difference, and there was no benefit even in the children whose pH probe was abnormal. What did change was harm: children on lansoprazole had more upper respiratory infections, sore throats and bronchitis.

The NICE guideline summarises the position in one sentence: reflux is more common in children with asthma, but it has not been shown to cause or worsen it. When asthma is poorly controlled, the right questions are about adherence, inhaler technique, triggers and whether the diagnosis is correct, not whether to start a proton pump inhibitor.

Part four. Infants

The same lesson applies to infants. In a trial by Orenstein and colleagues published in 2009, one hundred and sixty-two infants with symptoms attributed to reflux, including crying, irritability and cough, were randomly assigned to lansoprazole or placebo. Fifty-four percent responded in both groups. In other words, half of these babies got better with time, whatever they were given. Any adverse event occurred in sixty-two percent on lansoprazole and forty-six percent on placebo, a difference that just missed statistical significance. But serious adverse events occurred in ten infants on lansoprazole and two on placebo, a significant

difference, and most of these were lower respiratory tract infections. The 2018 paediatric guideline therefore states that a proton pump inhibitor trial should not be used as a diagnostic test in infants.

Part five. Why acid suppression may harm the airway

Why should a drug that lowers stomach acid cause chest infections? Gastric acid is an antimicrobial barrier. It kills most of the bacteria that are swallowed with saliva and food. When a proton pump inhibitor raises the gastric pH, those bacteria survive and multiply. Rosen and colleagues studied one hundred and sixteen children with chronic cough, fifty-nine of whom were taking proton pump inhibitors, and sampled their gastric fluid, lung washings and throat. The amount of Streptococcus in the stomach rose with the dose of the proton pump inhibitor. Eight groups of bacteria were shared by the stomach and the lungs but absent from the throat, which suggests they reached the lungs from the stomach rather than from the mouth. Two of them, Lactococcus and Acinetobacter, were more abundant in children who had more full-column reflux.

Here is the key point to carry into practice. A proton pump inhibitor changes the pH of the refluxate, not its volume or frequency. In a child who aspirates refluxate, acid suppression can convert sterile acidic material into non-acidic material that carries bacteria. That is one reason why a proton pump inhibitor is not a harmless just-in-case prescription for a child with recurrent chest infections.

Part six. The red larynx

Laryngeal findings are the most common reason children are labelled with reflux. In adults, the Reflux Finding Score, introduced by Belafsky, grades features such as swelling below the vocal folds, redness, swelling of the arytenoids and thick mucus. The Reflux Symptom Index is an adult questionnaire about throat symptoms.

Rosen and colleagues tested the Reflux Finding Score in seventy-seven children with chronic cough who also had pH-impedance and upper endoscopy. The score did not correlate with any measure of reflux; every correlation coefficient was smaller than zero point one five, which is essentially none. Agreement between two observers scoring the same images was poor, with an intraclass correlation coefficient of zero point three two, where one would mean perfect agreement. The positive predictive value of an abnormal score was twenty-nine percent against the pH study and forty percent against impedance. And the mean score was twelve in children with oesophagitis and twelve in children without it. The 2018 paediatric guideline states plainly that the appearance of the airway does not correlate with pathological reflux. Swelling and redness of the larynx are common in children who cough for any reason, in children who have been intubated and in children with allergic rhinitis. A report of reflux changes at laryngoscopy is a description, not a

diagnosis.

Part seven. Salivary pepsin and other markers

Pepsin is made only in the stomach, so in theory pepsin found in saliva or the airway should prove that stomach contents have travelled upwards. Commercial salivary pepsin tests are marketed directly to families. Dy and colleagues tested salivary pepsin against pH-impedance in fifty children. Forty-two percent were pepsin-positive. Against the reference test, the sensitivity of pepsin was forty-two percent, the specificity fifty-eight percent and the overall accuracy fifty percent, which is the accuracy of tossing a coin. A high cut-off, above seventy-four nanograms per millilitre, made a positive result fully specific but caught only seventeen percent of children with reflux. There was no correlation between pepsin concentration and the number of reflux episodes. The 2018 guideline recommends that salivary pepsin should not be used to diagnose reflux in infants and children, and more generally that currently available extraesophageal biomarkers should not be used.

The lipid-laden macrophage index is another marker. It counts macrophages in lung washings that contain fat droplets, on the idea that the fat comes from aspirated milk or food. Rosen and colleagues found that the index did not correlate with acid, non-acid or full-column reflux, or with oesophagitis, and concluded that it lacks the specificity to diagnose reflux-related lung disease. The fat in those cells can come from aspirated food during swallowing, from the lungs' own surfactant, from intravenous lipid infusions and from breakdown of cells. The guideline says the index should not be used for diagnosis.

What then is bronchoscopy for in a coughing child? Its value lies in the alternative diagnosis it can find: softening of the airway walls called tracheomalacia or bronchomalacia, a laryngeal cleft, compression of the airway by a vessel, a foreign body, or a bacterial bronchitis shown by neutrophils and a high bacterial load in the lavage. Ask the bronchoscopist to look for those things, and send the lavage for culture and cell count.

Part eight. Proving the link with pH-impedance

The one test that can relate an airway symptom to a reflux event is pH-impedance monitoring with symptoms marked as they happen. The 2018 guideline restricts its use to correlating persistent troublesome symptoms with reflux events, clarifying the role of reflux in extraesophageal signs and symptoms, judging whether acid suppression is working, and distinguishing the different forms of non-erosive reflux. NICE adds specific indications: suspected recurrent aspiration pneumonia, unexplained apnoeas, unexplained upper airway inflammation and frequent otitis media.

To get a useful answer, a few rules help. Study the child off acid suppression when the question is whether this is reflux at all, and on treatment when the question is whether

treatment is working. Mark each cough in real time, ideally with acoustic or pressure-based cough detection rather than relying on a diary. Look at the total burden of reflux, at full-column events, and at the symptom association statistics. The Lyon Consensus suggests that a symptom association probability above ninety-five percent, or a symptom index above fifty percent, increases confidence that reflux is causing the symptom, using a two-minute window. Remember that paediatric normal values are limited. And read the order of events, because cough followed by reflux can masquerade as an association. Objective testing is essential before any antireflux operation for an airway indication.

Part nine. Aspiration during swallowing

In infants and in children with neurological impairment, the most important mimic of reflux is aspiration during swallowing, called oropharyngeal aspiration. Here the food or drink goes into the airway on the way down, not on the way back up, and acid suppression is irrelevant.

Duncan and colleagues studied four hundred and twelve children under two years of age referred for a videofluoroscopic swallow study, which is a moving X-ray of swallowing with barium in different consistencies. No presenting symptom, including cough, choking or congestion, predicted who was aspirating. A clinical feeding evaluation at the bedside detected only forty-four percent of aspiration, and a reassuring bedside evaluation delayed the swallow study by about twenty-eight days. Aspiration is often silent, without any cough. The treatment is thickening of feeds, paced feeding and positioning, guided by a speech and feeding therapist. Thickening is usually temporary: in a study by Wolter and colleagues, thirty-nine of fifty infants, seventy-eight percent, were fully weaned back to thin liquids. So when a baby coughs with feeds, the first test is a videofluoroscopic swallow study, not a proton pump inhibitor.

Part ten. The aerodigestive clinic

Children with complex airway and feeding problems often have more than one cause for their symptoms. Aerodigestive programmes bring together gastroenterology, pulmonology, otolaryngology and speech and feeding therapy, usually with nutrition support and a nurse coordinator. Their signature procedure is triple endoscopy under a single anaesthetic: laryngoscopy and bronchoscopy with lavage, upper gastrointestinal endoscopy with oesophageal biopsies, and placement of a pH-impedance catheter, often combined with a swallow study around the same visit. A Delphi consensus published by Boesch and colleagues in 2018 reached agreement on thirty-three of thirty-six items covering the definition, the disciplines, the procedures and the outcomes of such programmes. At Johns Hopkins, a programme of one hundred and twenty-five children reduced the number of anaesthetics by forty-one percent, avoiding one hundred and nine anaesthetic episodes, and

reduced the number of separate visits for families.
Why does this work? Because seeing the larynx, the bronchi, the oesophagus and the swallow together allows the team to find a laryngeal cleft, eosinophilic oesophagitis or tracheomalacia that would otherwise stay hidden behind a reflux label. Children worth referring include those with cough and aspiration, suspected cleft, eosinophilic oesophagitis or airway malacia, children with neurological impairment, and children with repaired oesophageal atresia.

Part eleven. A practical pathway and the guidelines

A practical sequence for a child with chronic cough, wheeze or hoarseness that someone has attributed to reflux goes like this. Start with a paediatric cough work-up: decide whether the cough is wet or dry, obtain a chest X-ray and spirometry where the child is old enough, and treat a wet cough lasting more than four weeks with antibiotics for protracted bacterial bronchitis. Next, ask about typical reflux symptoms such as heartburn, regurgitation and vomiting; if they are present, a four to eight week trial of a proton pump inhibitor is reasonable, followed by reassessment, and the drug should be stopped if there is no response. If the child coughs with feeds, has neurological impairment or a wet voice, arrange a videofluoroscopic swallow study before any reflux test. If proving reflux would change management, arrange pH-impedance off treatment with cough marked. And if symptoms persist, look again with aerodigestive endoscopy for a cleft, malacia, eosinophilic oesophagitis or a foreign body.
Hoarseness deserves a separate word. In children, hoarseness is most often caused by vocal fold nodules from voice use, followed by allergy, infection and structural lesions. NICE advises against investigating or treating for reflux when cough, hoarseness or a single pneumonia is the only feature and there is no overt regurgitation. The referral to an ear, nose and throat surgeon is for a look at the vocal folds, not for a verdict on reflux.
How do the guidelines compare? On empiric proton pump inhibitors for isolated airway symptoms, they are aligned: the paediatric gastroenterology guideline says a trial should not be used as a diagnostic test, the chest physicians say acid suppression should not be used solely for cough, NICE says do not treat, and the adult gastroenterology societies say test before treating when typical symptoms are absent. On a short trial when typical reflux symptoms are also present, they align on four to eight weeks with reassessment. On salivary pepsin, the lipid-laden macrophage index and laryngoscopy as diagnostic tools, they align: none should be used to diagnose reflux. They differ on up-front physiological testing. Adult guidance, including the Lyon Consensus and the American Gastroenterological Association update, now favours testing first because it is more cost-effective than empiric therapy, while the paediatric guideline restricts pH-impedance to defined indications and reaches a similar place mainly by forbidding empiric treatment, with fewer validated normal values to

test against, especially in infants. On asthma they align that reflux treatment is not indicated without typical symptoms. And on surgery for airway symptoms, the paediatric guideline requires objective testing, while adult guidance discourages surgery after a failed response to proton pump inhibitors.

Part twelve. Traps, numbers and decisions

Several traps recur in practice. The first is escalation: the proton pump inhibitor failed, so the dose is doubled, when non-response is itself information. The second is accepting a red larynx or posterior laryngitis as proof. The third is starting treatment because a salivary pepsin kit was positive. The fourth is blaming a feeding-related cough on reflux and missing aspiration during swallowing. The fifth is leaving a child with a wet cough on a proton pump inhibitor for months when antibiotics for bacterial bronchitis were the answer. And the biggest is leaving a child on open-ended acid suppression for an airway symptom that never responded. Every proton pump inhibitor started for an extraesophageal indication needs a stop date and a defined outcome.

To bring the numbers together: almost eighty-four percent of coughs in coughing children are independent of reflux, and cough provokes reflux as often as the reverse. The chest physicians recommend against acid suppression solely for chronic cough. In the asthma trial, forty-three percent of children had silent abnormal acid exposure, and lansoprazole did not improve control but did add infections. In infants, fifty-four percent responded on lansoprazole and on placebo alike, with ten serious adverse events against two. The Cochrane review found adverse events with an odds ratio of about five and a half and a number needed to harm of eleven. Salivary pepsin has a sensitivity of forty-two percent, a specificity of fifty-eight percent and an accuracy of fifty percent. The Reflux Finding Score has a positive predictive value of twenty-nine percent against pH and poor agreement between observers. The lipid-laden macrophage index does not track reflux. A bedside feeding evaluation detects only forty-four percent of aspiration. And aerodigestive programmes reduce anaesthetics by about forty percent while finding the real cause behind a reflux label.

Consider five common situations. For the six-year-old girl from the start of this chapter, with a wet cough for four months, a red larynx and no response to eight weeks of omeprazole, stop the proton pump inhibitor, obtain a chest X-ray and treat for protracted bacterial bronchitis, because cough alone is not reflux and a wet cough usually has a bacterial cause. For a ten-year-old with poorly controlled asthma and no reflux symptoms, check adherence, inhaler technique and triggers, and do not start a proton pump inhibitor. For a four-month-old who coughs and splutters with every feed, arrange a videofluoroscopic swallow study, because aspiration during swallowing is likely and the bedside assessment misses about half of it. For a thirteen-year-old with heartburn and a cough at night, a four to

eight week trial of a proton pump inhibitor is reasonable, with reassessment and a stop if there is no response. And for a child with neurological impairment and recurrent pneumonia whose family has been offered a fundoplication, arrange an aerodigestive work-up with a swallow study, pH-impedance and bronchoscopy, so that aspiration during swallowing is separated from aspiration of reflux before any operation.

What should you do differently in clinic? Stop thinking of a proton pump inhibitor as a diagnostic test for airway symptoms. When one has already been started, ask what outcome it was meant to change and stop it if that outcome has not changed in four to eight weeks. Treat a wet cough as a chest problem first. Ask about feeding, and order a swallow study early in infants and in neurologically impaired children. Do not order salivary pepsin and do not accept a laryngeal appearance as proof. Reserve pH-impedance, with cough marked, for the child in whom the answer would change management. And for the complex child, use or build an aerodigestive pathway, because in these children the cause of the cough is rarely just reflux.